SIMATS ENGINEERING
Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO4 AT2 - Big Data Industry Problem Solving Guided Tool Workbook

Course Code	CSA15	**Slot**	B
Assessment	CO4 AT2	**Bloom Level**	BL4 - Analyze
Faculty	Dr C. Rajesh Babu	**Employee ID**	SSETSCS415
Mode	Individual guided tool-based workbook	**Submission**	GitHub repository link through Google Classroom
End Evidence	Workbook + screenshots + diagrams + dashboard/mockup	**Marks**	30

Student Details

Student Name		**Register Number**	
Class / Section		**Slot**	B
Selected Scenario		**Submission Date**	
GitHub Link		**Google Classroom Status**	

Workbook Purpose

What this workbook does
This workbook guides students to use real software tools and official technical references to derive a big-data industry solution. Students should learn from the tool, observe patterns, record evidence and then write their own technical explanation.

- Do not copy full text from websites or AI tools.
- Use official references to justify architecture, processing and security decisions.
- Screenshots are evidence; explanations are the actual assessment answer.

Scenario Bank

Instruction
Faculty may assign one scenario or ask students to complete more than one. If no specific scenario is assigned, select one scenario and complete all tool-guided sections for that scenario.

No.	Scenario	Data Context
1	**Healthcare Emergency Analytics**	Emergency department data, ambulance updates, bed availability, alerts and treatment delays.
2	**E-Commerce Customer Behavior Intelligence**	Clickstream events, cart abandonment, payment failure, campaign traffic and recommendation impact.
3	**Smart Agriculture Advisory Platform**	Weather, soil sensors, crop advisories, pest reports, market prices and farmer app interactions.
4	**Cybersecurity Threat Monitoring**	Authentication logs, firewall alerts, API access logs, failed login attempts and suspicious network events.
5	**Public Transport Demand Analytics**	Ticketing data, bus GPS, passenger count, route delay, peak-hour demand and crowding alerts.

Selected Scenario and Reason

Type response / paste screenshot / paste diagram here.

Guided Tool Module 1: Use Cloud Architecture Centers

Purpose: Use official cloud architecture references to identify suitable ingestion, storage, processing, analytics and security components for the selected scenario.

No.	Perform	Observe / Decide	Record in Workbook
1	Open AWS Architecture Center, Azure Architecture Center and Google Cloud Architecture Center in three browser tabs.	You are not copying a full architecture. You are learning common cloud patterns and service categories.	Paste the three page titles or URLs.
2	Search each site using terms from your scenario, such as healthcare analytics, streaming analytics, data lake, retail analytics, IoT analytics or security monitoring.	Notice which patterns repeatedly appear: ingestion, data lake, warehouse, processing, ML/analytics, visualization and security.	Record at least one useful reference from each cloud provider.
3	Identify the services/components mentioned in the reference diagrams.	Separate service category from vendor-specific service name. Example: object storage is the category; Amazon S3/Azure Blob Storage/Cloud Storage are service examples.	Complete the tool-category table.
4	Compare the three references.	Choose the most suitable architecture pattern for your scenario and explain why.	Write a short justification.
5	Draw your own provider-neutral architecture in draw.io.	Your diagram must show data sources -> ingestion -> storage -> processing -> analytics/dashboard -> users, with security controls.	Paste the diagram screenshot/link.

Cloud Provider	Reference Page / URL	Useful Pattern Found	Useful Service Categories

Cloud Architecture Justification

Type response / paste screenshot / paste diagram here.

Architecture Diagram Evidence

Type response / paste screenshot / paste diagram here.

Guided Tool Module 2: Use Hadoop and Spark Documentation

Purpose: Decide whether your problem needs batch processing, streaming processing or hybrid processing by reading official Hadoop/Spark documentation.

No.	Perform	Observe / Decide	Record in Workbook
1	Open the Apache Hadoop MapReduce tutorial.	MapReduce is suitable when large stored data can be processed in batches, such as daily logs or historical transactions.	Write one sentence on what mapper and reducer mean.
2	Open the Apache Spark Structured Streaming guide.	Structured Streaming is suitable when new events continuously arrive, such as sensors, clickstream, live alerts or location updates.	Write one sentence on what streaming source, processing and sink mean.
3	Classify your scenario data into batch data and streaming data.	Historical records normally become batch data; live sensor/log/API updates normally become streaming data.	Complete the batch-vs-stream table.
4	Choose one processing approach for your scenario: Hadoop/MapReduce, Spark batch, Spark streaming or hybrid.	The decision must match the data velocity and decision urgency.	Write your processing decision.
5	Prepare a simple pseudo-flow, not full code.	Example: Mapper reads log line -> emits hour, count. Reducer sums counts. Dashboard shows peak hour.	Write pseudo-flow or paste diagram.

Data Item	Batch / Streaming	Why	Suggested Framework

Processing Framework Decision

Type response / paste screenshot / paste diagram here.

MapReduce / Spark Pseudo-Flow

Type response / paste screenshot / paste diagram here.

Guided Tool Module 3: Threat Modeling with OWASP Threat Dragon / Microsoft Tool

Purpose: Identify privacy and security risks in the big-data solution before proposing controls.

Tool Choice
Use OWASP Threat Dragon if you need a web/desktop cross-platform threat modeling tool. Use Microsoft Threat Modeling Tool when a Windows system is available. The required evidence is the threat model diagram and threat-control table.

No.	Perform	Observe / Decide	Record in Workbook
1	Open OWASP Threat Dragon or Microsoft Threat Modeling Tool.	A threat model is a security diagram that shows components, data flows, trust boundaries and possible threats.	Write which tool you used.
2	Create a new threat model named with your scenario.	Use a clear title such as CO4_AT2_Healthcare_Emergency_Analytic s.	Paste model name.
3	Add external entities.	External entities are users/systems outside your platform: patient, customer, sensor, payment gateway, bus GPS device, public API.	List at least two external entities.
4	Add processes and data stores.	Processes transform data; data stores hold data. Example: ingestion API, stream processor, data lake, warehouse, dashboard.	List processes and stores.
5	Draw data flows and trust boundaries.	Trust boundaries mark where security assumptions change: public internet to cloud, app to API, processing layer to storage.	Paste screenshot of DFD.
6	Identify threats using STRIDE thinking.	Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service and Elevation of Privilege help structure risks.	Complete the threat table.
7	Write mitigations.	Controls may include authentication, authorization, encryption, masking, audit logs, rate limits, backup and monitoring.	Map every threat to one control.

Component / Data Flow	Threat	Impact	Mitigation

Threat Model Diagram Evidence

Type response / paste screenshot / paste diagram here.

Guided Tool Module 4: Dashboard or Analytics Output Mockup

Purpose: Show how raw big data becomes useful decisions through a dashboard, alert, forecast, recommendation or report.

Beginner-Friendly Choice
Use any one: Power BI, Looker Studio, Tableau Public or Canva. If login/install is difficult, Canva dashboard mockup is acceptable. If data connection is possible, Power BI/Looker Studio/Tableau Public is stronger.

No.	Perform	Observe / Decide	Record in Workbook
1	Prepare a small sample dataset in Excel or Google Sheets with 10-20 rows.	Columns should match your scenario. Example: timestamp, user/location, event type, value, status, risk level.	Paste column names.
2	Choose the dashboard tool.	Power BI/Tableau/Looker are analytics tools; Canva is acceptable for visual mockup when live data connection is blocked.	Write chosen tool and reason.
3	Import or recreate the sample data.	In Power BI/Looker/Tableau, connect data. In Canva, manually create charts from sample values.	Paste screenshot of data/chart setup.
4	Create at least three visual elements.	Use KPI card, bar chart, line chart, table, map, alert count or pie/donut chart as appropriate.	List the visuals.
5	Add decision labels.	A dashboard is useful only if it supports a decision: allocate beds, detect fraud, reroute buses, restock products, warn farmers.	Write decision supported by each visual.
6	Export/share screenshot.	Paste the final dashboard screenshot or public/report link if available.	Paste evidence.

Dataset Column	Meaning	Example Value

Visual Element	Metric Shown	Decision Supported

Dashboard / Analytics Output Evidence

Type response / paste screenshot / paste diagram here.

Final CO4 AT2 Solution Sections

Final Assembly
After completing the guided tool modules, combine the evidence into the final solution. The final answer must show technical reasoning, not only screenshots.

1. Problem Understanding

What to include: Identify stakeholders, decisions required, data produced, expected analytics output and urgency.

Student Response

Type response / paste screenshot / paste diagram here.

2. 5Vs Mapping

What to include: Map Volume, Velocity, Variety, Veracity and Value using scenario evidence.

Student Response

Type response / paste screenshot / paste diagram here.

3. Big Data Architecture

What to include: Paste final draw.io architecture showing sources, ingestion, storage, processing, analytics/security and users.

Student Response

Type response / paste screenshot / paste diagram here.

4. Data Flow Pipeline

What to include: Explain how raw data moves, gets cleaned, stored, processed and converted into dashboard/alert/decision.

Student Response

Type response / paste screenshot / paste diagram here.

5. Tool and Technology Justification

What to include: Use cloud architecture references and Hadoop/Spark documentation to justify chosen tools.

Student Response

Type response / paste screenshot / paste diagram here.

6. Security and Privacy Strategy

What to include: Use threat model output to explain threats and controls.

Student Response

Type response / paste screenshot / paste diagram here.

7. Challenges and Solutions

What to include: Write at least four big-data implementation challenges and one practical solution for each.

Student Response

Type response / paste screenshot / paste diagram here.

8. Business Outcome

What to include: State measurable outcome: faster decision, reduced fraud, lower waiting time, improved prediction, lower downtime or better user experience.

Student Response

Type response / paste screenshot / paste diagram here.

Rubric

Criteria	Marks	Evaluation Focus	Evidence
Problem and 5Vs analysis	6	Scenario-specific analysis and correct big-data reasoning	Workbook tables
Architecture and data flow	6	Correct pipeline, cloud components and flow clarity	draw.io diagram
Processing framework decision	5	Batch/streaming/hybrid justification using Hadoop/Spark concepts	Decision table
Security and privacy	5	Threat model, risks and controls	Threat model table/screenshot
Dashboard/output design	4	Useful decision-oriented analytics output	Dashboard/mockup
Clarity and submission quality	4	Organized, original, complete evidence	GitHub/GCR

Submission Checklist

- Student details and selected scenario are filled.
- Cloud architecture references are recorded.
- Hadoop/Spark processing decision is completed.
- Threat model diagram/table is included.
- Dashboard/mockup screenshot or link is included.
- Final solution sections are completed.
- DOCX/PDF and supporting files are uploaded to GitHub.
- GitHub repository link is submitted through Google Classroom.

Official Reference Links

AWS Architecture Center: https://aws.amazon.com/architecture/

Azure Architecture Center: https://learn.microsoft.com/en-us/azure/architecture/

Google Cloud Architecture Center: https://docs.cloud.google.com/architecture

Apache Hadoop MapReduce Tutorial: https://hadoop.apache.org/docs/stable/hadoop-mapreduce-client/hadoop-mapreduce-client-core/MapReduceTutorial.html

Apache Spark Structured Streaming: https://spark.apache.org/docs/latest/streaming/index.html

Microsoft Threat Modeling Tool releases: https://learn.microsoft.com/en-us/azure/security/develop/threat-modeling-tool-releases

OWASP Threat Dragon: https://owasp.org/www-project-threat-dragon/

Power BI documentation: https://learn.microsoft.com/en-us/power-bi/

Looker Studio create report: https://docs.cloud.google.com/data-studio/create-a-report

Tableau Public: https://www.tableau.com/community/public

Canva Charts: https://www.canva.com/help/charts/